

2(a)	glycosidic ; I any qualification of glycosidic, e.g. β ,1–4	1
2(b)(i)	hydrolysis ;	1
2(b)(ii)	<i>any two from</i> (lactose) lowers the water potential (in the lumen of the intestine) / AW ; reduces the water potential gradient (between the intestine and the blood) ; <u>osmosis</u> occurs ; (must be in the correct context)	2
2(c)(i)	<i>any two from</i> pack more beads into the column ; slows the passage of milk through the column ; more time for enzyme to be exposed to substrate ; (small beads have a) larger surface area to volume ratio ; rate of reaction will be faster / AW ; A faster reactions will take less time to collect results (at each temperature) ; fast(er), rate of diffusion of, substrate / product ;	2
2(c)(ii)	<i>any three from</i> both have same activity between 0 and 20 °C ; A described with values from Fig. 2.2 between 20 °C and 40 °C, the activity of F increases more steeply than I ; F , has a lower optimum temperature / peaks at a lower temperature / AW, than I ; A optimum temperature, F is 40 °C v I is 50 °C at 70 °C, no activity for F but there is activity for I ; A 4% for I and 0% for F A denatured for 0% above the optimum temperature I has a steeper decrease of activity ; I is active over a greater range of temperatures (than F) ;	3